

WITHAQ

Function-Aware Device Containment and Data Revocation for Trusted Workflow Recovery

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Introduction

Protect the function, then control its data.

Connected devices must remain useful during security incidents. Full isolation can interrupt monitoring and alerts. Operational-security guidance recognizes availability needs [1]; device policies describe expected communications [2].

Research gap

Which device functions can continue while suspect data and its derivatives are restricted? WITHAQ links per-function containment to data revocation and verified recovery.

Objective

Test whether bounded containment can block modeled attack paths while preserving every critical-function contract, then recover outputs from approved sources.

Methodology

Proposed end-to-end response

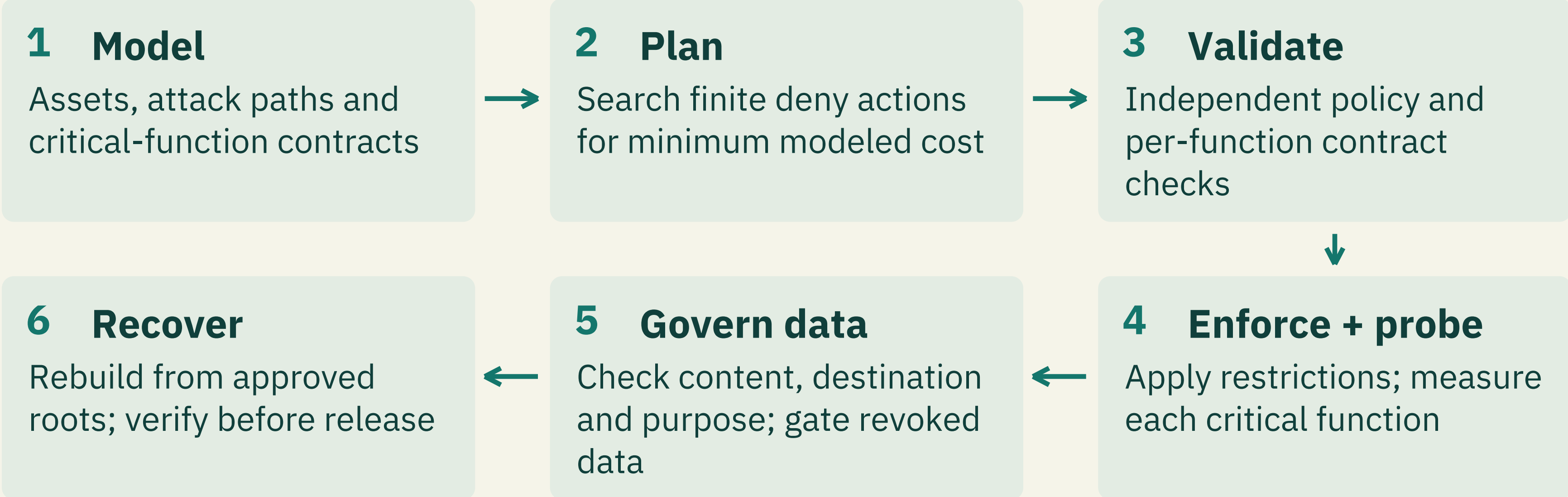


Figure 1. Proposed control sequence. Reject invalid plans; keep unknown states explicit. No language-model approval authorizes enforcement.

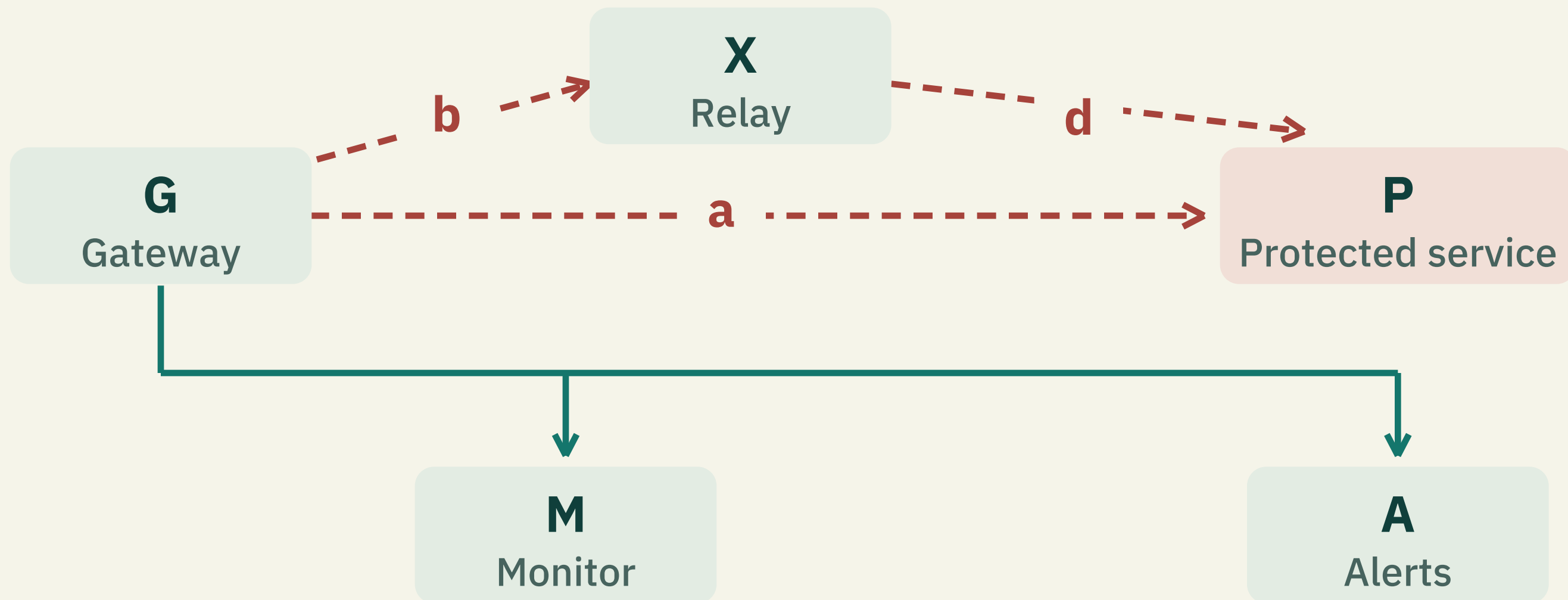
Planned limits: sample age ≤ 2 s; each alert delivered within 2 s.

Python enumerates all finite action sets. Planned lab: Linux namespaces, nftables, 1 Hz synthetic streams and independent probes. Compare 6 scenario families x 10 seeded configurations x 5 policies (300 settings) [5]; log every contract failure and unknown state.

Results

ILLUSTRATIVE MODEL CHECK

No operational experiments completed.



c: isolate all G links

Figure 2. Toy topology [5]. Red: attack routes; green: required functions. Actions a, b and d block their labeled links; c isolates G.

Action set	Cost	Paths blocked	Functions kept	Decision
{a}	1	1 / 2	2 / 2	Reject
{c}	1	2 / 2	0 / 2	Reject
{a, b}	2	2 / 2	2 / 2	Accept

Table 1. Selected sets from all 16 combinations. Three are feasible; {a, b} has minimum cost in this toy model (a = b = c = 1; d = 2).

Innovation

Proposed novelty: join function-aware containment, disclosure control and lineage-based recovery in one verifiable response.

01 Per-function acceptance

Reject any plan that breaks a critical contract, even if other functions remain available.

02 Source-impact control

Trace dependencies [3]; guard reads, publication and dispatch after a source is revoked.

03 Verified recovery

Rebuild affected outputs from approved roots. Continue independent branches only after verification.

Prior work includes device access policy [2], provenance [3] and agent capability controls [4]. The contribution is their proposed response contract; global novelty and superiority are not established.

Conclusion

CURRENT STAGE: PROPOSED ARCHITECTURE

Design output: a response architecture and a finite check with 3 feasible action sets out of 16. The example verifies a logical choice; operational performance remains untested.

Expected value

Fewer avoidable service interruptions and controlled recovery for device operators. These benefits require lab validation.

Limits: connectivity does not prove sensor truth. Already-dispatched data and external model weights remain outside scope.

Future Work & References

Next 6-8 weeks (planning estimate) [5]

- Build lab, planner and independent validator.
- Integrate enforcement and data guards.
- Compare policies; test races and recovery.

References [1-5]



Scan to read the complete bibliography as text, including public source links. No sign-in required.

Planned measures: attack denial, per-function failures, response latency and recovery validity.